QUIZ4-Page1-4

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

11 / 14 pts

Question 1

Q1

7 / 7 pts

+ 0 pts Incorrect

✔ **+ 7 pts** Correct Answer

+ 3 pts Part (a) is correct.

+ 1.5 pts Correctly computed $\|T(f) - T(g)\|_\infty$

+ 1.5 pts Correctly observe the value of K which gives the contraction.

+ 4 pts Part (b) is correct.

Question 2

Q2

4 / 4 pts

+ 0 pts Incorrect

✔ **+ 4 pts** Fully Correct.

+ 0.5 pts Construct a function $F_n(g) = \frac{1}{n}\sum_{i=1}^{n} g(x_i)$

+ 1 pt Approximate $F_n(g)$ by polynomial.

+ 0.5 pts Again construct the function $F_n(P_k) = \frac{1}{n}\sum_{i=1}^{n} P_k(x_i)$

+ 1 pt Limit of $F_n(P_k)$ is finite.

+ 1 pt Concluding the result.

Question 3

Q3

0 / 3 pts

✔ **+ 0 pts** Incorrect

+ 3 pts Fully Correct.

+ 1.5 pts Construct a new function.

+ 1 pt Apply the Dini theorem.

+ 0.5 pts Concluding the uniform convergence.

ANALYSIS - I : MTH 301 : QUIZ : 2025-26

Name: MATIN PREETAM BHOJWANI Roll No: 241211

- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only. Maximum marks: 25 Time: 18:30 - 19:45 .

1. Suppose that $F : \mathbb{R}^2 \to \mathbb{R}$ is continuous and Lipschitz in its second variable, i.e. $|F(r, s) - F(r, t)| \le K|s - t|$ for all $r \in \mathbb{R}$. Define $T : C([0, 1]) \to C([0, 1])$ by $(Tf)(x) = \int_0^x F(t, f(t))\, dt$.

   (a) Justify whether T is always a strict contraction or not.

   (b) Show that T maps bounded sets into equicontinuous sets.

[3 + 4]

$F : \mathbb{R}^2 \to \mathbb{R}$ is continuous.
and Lipschitz in its second variable.

$|F(r,s) - F(r,t)| \le K|s-t|$

$T: C([0,1]) \to C([0,1])$

$(Tf)(x) = \int_0^x F(t, f(t))\, dt$.

(a). for T to be strict contraction.

$\|T(f) - T(g)\|_\infty \le \alpha \|f-g\|_\infty$

now $\left\| \int_0^x F(t,f(t))\,dt - \int_0^x F(t,g(t))\,dt \right\|_\infty$

$\le \left\| \int_0^x F(t,f(t)) - F(t,g(t))\, dt \right\|$

$\le \sup_{x \in [0,1]} \left| \int_0^x F(t,f(t)) - F(t,g(t))\, dt \right|$

$\le \sup_{x\in[0,1]} \int_0^x |F(t,f(t)) - F(t,g(t))|\,dt$

$\le \sup \int_0^x K\,|f(t) - g(t)|\,dt$

$\le K \sup_{x \in [0,1]} \int_0^x |f(t) - g(t)|\,dt$

$= K\|f-g\|_\infty$

now this depends on value of K and for strict contraction $0 < K < 1$ which is not given. hence it is not always a strict contraction.

(b) Let $A \subset C([0,1])$ that is bounded.
$\exists M$ s.t. $\|f\|_\infty < M$ $\forall f \in A$.

now consider $g \in T(A)$
where $g(x) = T(f) = \int_0^x F(t, f(t))\,dt$

for equicontinuous take $\varepsilon > 0$ $\exists \delta > 0$
$|g(y) - g(x)| < \varepsilon$ when $|y - x| < \delta$

now $|g(y) - g(x)| = \left| \int_x^y F(t, f(t))\,dt \right|$

w.l.o.g assume $y \ge x$

$|g(y) - g(x)| = \left| \int_x^y F(t, f(t))\,dt \right|$

$\le \int_x^y |F(t, f(t))|\,dt$

$\le$ ~~$\int_x^y F(t,f(t))\,dt$~~ (as ~~the~~ grad of the)

now we know $|f(t)| < M$ $\forall t$

$\Rightarrow (t, f(t)) \in [0,1] \times [-M, M]$

this is a compact ~~space~~ set in $\mathbb{R}^2$. $\exists$ $F(t, f(t))$ attains a maximum denote it by L.

$\le \int_x^y |L|\,dt = |L|(y-x)$

now
$|g(y) - g(x)| \le |L|\,|y-x|$

take $\delta = \frac{\varepsilon}{L}$

Let $|y-x| < \delta$

then $|g(y) - g(x)|$

$\le |L|\,|y-x|$

$\le L\,\delta$

$\le L\,\frac{\varepsilon}{L} = \varepsilon$

$\Rightarrow |g(y) - g(x)| \le \varepsilon$ when $|y-x| < \delta$

independent of g & y, x

hence equicontinuous

········· Rough Work ·········

Σf_n. $f_n($ there are. polynomials that

$f_n(x_m)$

x ; $\|f_n - g\|$

to be equicontinuous.

$|$ ~~$g(x) - f(x)$~~

$|g(y) - g(x)| < \varepsilon$.

whenever.

$\sup_{x \in [0,1]} \int_0 \ldots \Big|$ $\int_0 F(t,f(t))dt\Big|$